

1.9 Pre-Quiz: Arithmetic and Geometric Sequences — Solutions

Review set, not scored: worked solutions only, no marks. Calculator allowed. Answers are exact or correct to three significant figures.

1. 12, 19, 26, 33, ...

(a) $d = 19 - 12 = 7$

(b) $u_n = u_1 + (n - 1)d = 12 + 7(n - 1) = 7n + 5$

Check: $u_1 = 7 + 5 = 12$.

(c) $u_{40} = 7(40) + 5 = 285$

(d) $7n + 5 = 250 \Rightarrow 7n = 245 \Rightarrow n = 35$

n is a whole number, so 250 is the 35th term.

(e) $7n + 5 = 300 \Rightarrow 7n = 295 \Rightarrow n = 42.14\dots$

n is not a whole number, so 300 is not a term.

Equivalently: $u_{42} = 299$ and $u_{43} = 306$, and 300 lies between them.

2. $u_5 = 31$, $u_{12} = 3$.

(a) From u_5 to u_{12} is 7 steps of d : $7d = 3 - 31 = -28 \Rightarrow d = -4$

$u_5 = u_1 + 4d \Rightarrow 31 = u_1 - 16 \Rightarrow u_1 = 47$

Or solve the simultaneous equations $u_1 + 4d = 31$, $u_1 + 11d = 3$ on the GDC.

(b) $u_n = 47 - 4(n - 1) = 51 - 4n$

(c) $51 - 4n < 0 \Rightarrow n > 12.75$, so $n = 13$.

$u_{12} = 3$ and $u_{13} = -1$. The first negative term is -1 (the 13th term).

The question asks for the term, not n : the answer is -1 .

(d) Equal differences: $3k - (k + 2) = (4k + 1) - 3k$

$2k - 2 = k + 1 \Rightarrow k = 3$

First three terms: 5, 9, 13 ($d = 4$)

(e) $u_{20} = 5 + 19 \times 4 = 81$

3. (a)

- (i) 3, 7, 11, 15 arithmetic, $d = 4$
- (ii) 3, 6, 12, 24 geometric, $r = 2$
- (iii) 3, 4, 6, 9 neither (differences 1, 2, 3; ratios 1.33, 1.5, 1.5)
- (iv) 80, 20, 5, 1.25 geometric, $r = \frac{1}{4} = 0.25$

Test arithmetic by subtracting consecutive terms, geometric by dividing. In (iii) neither the differences nor the ratios are all equal.

(b) $r = \frac{9}{6} = 1.5$ (check: $\frac{13.5}{9} = 1.5$)

(c) $u_n = u_1 r^{n-1} = 6(1.5)^{n-1}$

(d) $u_{10} = 6(1.5)^9 = 230.66 \dots \approx 231$

The exponent is 9, not 10. $6(1.5)^{10} = 346$ is the 11th term.

(e) $6(1.5)^{n-1} > 1000$

$u_{13} = 778.5$ (< 1000); $u_{14} = 1167.7$ (> 1000)

$n = 14$

Or solve $1.5^{n-1} > 166.7$ on the GDC: $n - 1 > 12.6$, so $n = 14$. The answer is a whole number.

(f) From u_2 to u_5 is 3 steps of $\times r$: $r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$

$u_1 = \frac{u_2}{r} = \frac{12}{3} = 4$

Check: 4, 12, 36, 108, 324.

4. Job A: arithmetic, $u_1 = 42\,000$, $d = 1500$. Job B: geometric, $u_1 = 40\,000$, $r = 1.05$.

(a) Job A: $42\,000 + 2 \times 1500 = \$45\,000$

Job B: $40\,000 \times 1.05^2 = \$44\,100$

(b) Job A: $A_n = 42\,000 + 1500(n - 1) = 40\,500 + 1500n$

Job B: $B_n = 40\,000(1.05)^{n-1}$

- (c) Compare the two salaries year by year:

Year	Job A	Job B
4	\$46 500	\$46 305
5	\$48 000	\$48 620.25

Year 5

Use a GDC table of both expressions. Show the two adjacent years: year 4 is what shows that year 5 is the first.

(d) $3000 \times 1.03^8 = \$3800.31$ (\$3800 to 3 s.f.)

TVM solver: $N = 8$, $I\% = 3$, $PV = -3000$, $PMT = 0$, $P/Y = C/Y = 1$.

(e) Real value = $\frac{3800.31}{1.02^8} = \3243.53 (\$3240 to 3 s.f.)

The real-interest-rate method is also accepted: $3 - 2 = 1\%$ per year, $3000 \times 1.01^8 = \$3248.57$ (\$3250 to 3 s.f.).

The real value must be less than the answer to (d). Multiplying by 1.02^8 is the most common error.